

JUXTAGLOMERULAR APPARATUS (JGA)

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- **Definition** : A specialized microscopic structure situated near the glomerulus of each nephron (*Juxta* = *near*).
- **Components** : Formed by **3 principal cellular elements**:
 - a. Macula Densa
 - b. Extraglomerular Mesangial Cells
 - c. Juxtaglomerular (JG) Cells

COMPONENTS OF THE JGA

1. Macula Densa

- **Location** : Located in the terminal portion of the **thick ascending limb of the loop of Henle** just before it opens into the distal convoluted tubule (DCT).
- **Position** : Lies in the acute angle between the **afferent and efferent arterioles** of the same nephron (in very close contact with the afferent arteriole).

- **Structure** : Formed by **tightly packed, specialized cuboidal/columnar epithelial cells** that act as chemoreceptors sensing NaCl concentration in tubular fluid.

2. Extraglomerular Mesangial Cells

- **Location** : Situated within the **triangular region** bounded by the macula densa, afferent arteriole, and efferent arteriole.
- **Synonyms** :
 - • Agranular cells
 - • Lacis cells
 - • Goormaghtigh cells

3. Juxtaglomerular (JG) Cells

- **Definition** : Specialized, modified vascular **smooth muscle cells** located primarily in the wall of the **afferent arteriole** just before it enters Bowman's capsule.
- **Location in Vessel Wall** : Present predominantly within the **tunica media** and **tunica adventitia** of the afferent arteriole.
- **Synonyms & Characteristics** :
 - Also called **Granular cells** because their cytoplasm contains prominent secretory granules storing **renin**.
 - Aggregation of these cells forms the **polar cushion / Polkissen**.

RELATED: GLOMERULAR (INTRAGLOMERULAR) MESANGIAL CELLS

- **Location** : Situated within the glomerular tuft between the **capillary loops**.
- **Key Functions** :—
 - **Structural Support** : Provide architectural support for the glomerular capillary loops.
 - **Regulation of GFR** : Exhibit **contractile properties**; contraction decreases the effective capillary surface area, thereby reducing GFR.
 - **Phagocytic Activity** : Phagocytose trapped immune complexes and debris to keep the filtration barrier clear.

- **Secretory Function** : Synthesize and secrete:
 - • Glomerular interstitial matrix (mesangial matrix)
 - • Prostaglandins
 - • Cytokines

STRUCTURAL SUMMARY

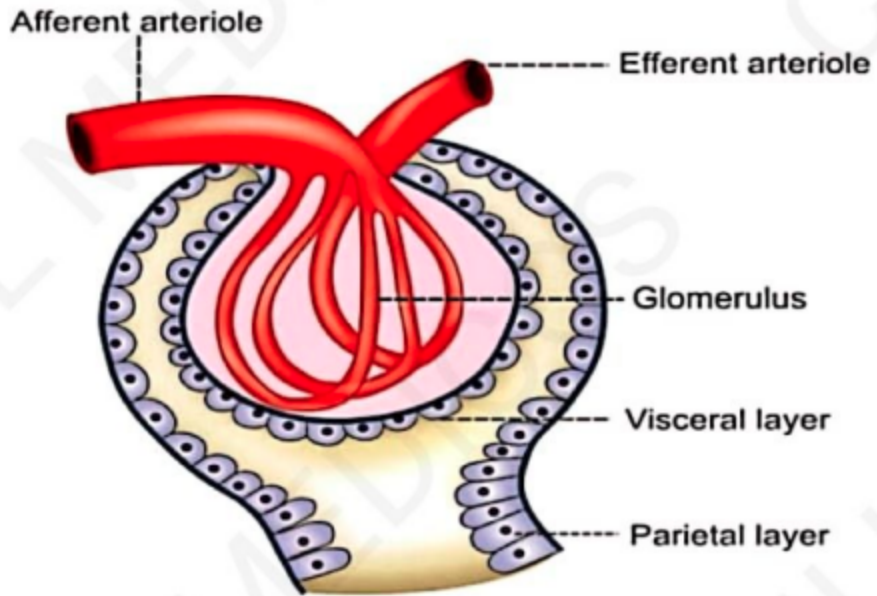


FIGURE 49.3: Renal corpuscle

Intraglomerular Component

Glomerular Mesangial Cells

- Supports capillary loops
- Contraction regulates GFR
- Phagocytic & secretory

Juxtaglomerular Apparatus (JGA)

Macula Densa

- Thick ascending limb / DCT
- Chemoreceptor for tubular NaCl

Extraglomerular Mesangial Cells

- Lacis / Goormaghtigh cells
- Fills triangular vascular space

Juxtaglomerular Cells

- Modified smooth muscle of Afferent Arteriole
- Granular cells (Polkissen)
 - Synthesize & secrete Renin

FUNCTIONS OF JUXTAGLOMERULAR APPARATUS

1. SECRETION OF HORMONES

The Juxtaglomerular (JG) apparatus secretes **2 principal hormones**:

- i. **Renin**
- ii. **Prostaglandins**

A. Renin

- **Site of Secretion** : Secreted by **Juxtaglomerular (JG) cells** (*granular cells of the afferent arteriole*).
- **Nature** : Peptide composed of **340 amino acids**.
- **Physiological Role** : Forms the **Renin-Angiotensin System (RAS)** in conjunction with angiotensins for the regulation and maintenance of systemic blood pressure (BP).
- **Stimuli for Renin Secretion** :—
 - a. Fall in arterial blood pressure.
 - b. ↓ Decrease in Extracellular Fluid (ECF) volume.
 - c. ↑ Increased sympathetic nerve activity.
 - d. ↓ Decreased tubular load of sodium (Na^+) and chloride (Cl^-) reaching the macula densa.

B. Prostaglandins

- **Site of Secretion** : Secreted by **Extraglomerular mesangial cells** (*Lacis / Goormaghtigh cells*).
- (*Note: Medullary interstitial cells [Type-I interstitial cells / MIC] also secrete prostaglandins.*)

2. SECRETION OF OTHER SUBSTANCES

- **Extraglomerular Mesangial Cells** : Secrete vital **cytokines**, including:
 - • **Interleukin-2 (IL-2)**
 - • **Tumor Necrosis Factor (TNF)**
- **Macula Densa** : Synthesizes and secretes **Thromboxane A_2 (TXA_2)**.

3. REGULATION OF GLOMERULAR BLOOD FLOW & GFR

Macula Densa senses luminal NaCl delivery



Activates **Tubuloglomerular Feedback (TGF)** mechanism



Regulates Glomerular Blood Flow (GBF) & Glomerular Filtration Rate (GFR)

FUNCTIONAL SUMMARY TABLE

Cellular Component	Secretory Products / Mediators	Primary Physiological Role
Juxtaglomerular Cells	Renin (340 AA peptide)	Initiates the RAS cascade to maintain systemic arterial pressure and fluid balance.
Extraglomerular Mesangial Cells	• Prostaglandins • Cytokines (IL-2, TNF)	Local vascular autoregulation and paracrine immune/inflammatory mediation.
Macula Densa	• Thromboxane A₂	Acts as the chemosensory hub for Tubuloglomerular Feedback , adjusting GFR and GBF.